

Evidence-preserving realist synthesis and verification

Edition: 23 September 2026. The fixed-corpus machine baseline is complete; human validation is pending. The separate evaluation-v2 attempt has zero completed responses because the provider rejected its first request for exhausted credit. Semantic-assertion contracts are implemented and tested, but semantic extraction under that extension has not run. This document distinguishes observed outputs, implemented controls and proposed work.

Research question and contribution

The research question is whether a controllable agent workflow can perform the analytic operations of a realist evidence synthesis and produce explanations that experts can inspect, challenge and compare with a published human synthesis. Richmond's clinical-reasoning review is a worked example. The contribution must be demonstrated through traceable analytic decisions and evaluation of the resulting explanations. A running interface, a dense graph or a high label-overlap score alone does not answer the research question.

The accepted RRE proposal describes researcher-controlled roles, an educational literature knowledge graph, graph-enhanced retrieval, domain adaptation and evaluation at module and system levels. Meetings distinguish two experiments: recovering the original included literature through searching/screening, and reconstructing the synthesis on the fixed included corpus. They also permit architectural adaptation while maintaining the methodological aim. These commitments are design requirements, not evidence that every proposed component already exists.^[1]

The immediate experiment uses the fixed 28-paper registry and explicitly reports document availability. It does not claim to repeat four subscription-database searches or reconstruct a complete historical screening population. Retrieval and screening require their own dated exports, search syntax, inclusion decisions and denominators. This separation prevents an apparent synthesis success from concealing a retrieval failure, or an artificial screening pool from being presented as the original review.

Evidence from related research

Source	Finding relevant to this project	Decision and boundary
RAMESES realist-synthesis reporting standards	Transparent explanation of selection, appraisal, extraction and iterative synthesis is central; findings include articulated inferences, not only extracted observations. ^[2]	Preserve the chain from source to inference and explain contingent recommendations. A report satisfying a checklist is not automatically a valid synthesis.
Dalkin et al., mechanism conceptualization	Separating intervention resources from participants' reasoning helps operationalize mechanisms; activation may vary continuously rather than operate as a simple binary switch. ^[3]	Represent resource and response separately. Do not treat an intervention label as a complete mechanism or a graph arrow as a measured causal effect.
Microsoft GraphRAG	Graph-based community summaries support corpus-level, query-focused summarization. ^[4]	Use graph-organized evidence summaries as an inspectable retrieval/synthesis component. Its summarization results do not establish realist validity or faithful causal reconstruction.
PaperQA2	Evaluates retrieval, cited synthesis and contradiction detection with task-specific human comparisons. ^[5]	Adopt explicit evidence access and separate evaluation tasks. Performance on its tasks cannot be transferred to this corpus or used to claim human equivalence here.
LatteReview	Modular agents support review workflows, but screening performance varies by dataset and decision threshold. ^[6]	Separate recall-sensitive screening from synthesis. Do not select high recall from one policy and high specificity from another and present them as one operating point.
Self-Refine	Iterative feedback and revision can improve outputs on tested tasks without parameter training. ^[7]	Permit bounded feedback loops, retain revisions and evaluate their effect. Prompt revision is not fine-tuning or RLHF.
Huang et al., self-correction study	Intrinsic self-correction without external feedback is unreliable in the reasoning settings studied. ^[8]	A second model opinion is a diagnostic signal. Returning to sources and human adjudication remain necessary.
SciFact	Scientific claim verification separates evidence retrieval, stance and supporting rationales. ^[9]	Record supporting, conflicting and insufficient evidence distinctly. Verification of an isolated claim does not establish a whole realist explanation.
FactScore	Evaluates support at the level of atomic factual claims rather than giving an entire answer one binary judgment. ^[10]	Check each component, qualification and citation. Add configuration-level comparison because individually correct fragments can still form an incorrect explanation.
Louvain/Leiden research and NetworkX implementation	Louvain partitions can be disconnected; Leiden addresses stronger connectivity properties. NetworkX exposes a seeded Louvain implementation. ^[11]	The current diagnostic implementation splits disconnected Louvain groups. It does not claim this is Leiden or that connectivity establishes a coherent theory.

These sources support specific design principles. They do not jointly prove that this implementation is effective. The scientific claim must come from the project's own measured comparison, including failures and expert judgments. A newer preprint describing multi-agent review automation was also located, but its abstract alone is insufficient to justify changing the architecture or claiming novelty relative to it; it remains a follow-up source rather than a result relied upon here.^[12]

Unit of analysis and evidence representation

A finding is represented as

```
f = (paper, study_family, C, R, M, O, direction, comparator, time, evidence_status, evidence_spans,
limitations).
```

c is the relevant context; *R* is the resource offered; *M* is the learner's response; *o* is an outcome. This is an operational representation of a realist configuration, not a structural equation that estimates a causal effect. Some fields may be unknown. A paper can illuminate only one part of an explanation, and a conceptual paper can describe a plausible resource or mechanism without an observed outcome.

Each response is labelled reported, author inference, model hypothesis, or not reported. Each outcome is labelled measured, reported perception, author inference, or not reported. These labels prevent an account of enjoyment from becoming an accuracy result and prevent a discussion-section explanation from becoming a measured mediator. Inference remains legitimate, but it must be inspectable.

The source identity includes the PDF hash, paper identifier, PDF page and character offsets. Evidence quotations retain the model-proposed text and, where located, the actual underlying source span. Exact matching is attempted first. Typography normalization permits whitespace, Unicode ligatures and line-end hyphenation, while preserving an offset mapping. It does not permit fuzzy semantic similarity, omission of a negation, replacement of an outcome or selective matching of a convenient substring.

S013 and S020 share a study-family identifier because one report analyzes material associated with the other. Counts of papers, extracted findings, theory statements and independent study families are separate quantities. A high number of extracted findings does not mean a large independent evidence base.

Precise algorithm

1. Freeze the experiment

Read the fixed registry, preserve its historical S001-S028 identifiers, and record hashes and availability. Correct the visibly erroneous Linn title/year without renumbering the registry. Snapshot the configuration and actual execution code identity. Store each API request, returned model identifier, raw response, parsed response, token use and estimated cost. Reuse a completed call only if its request hash matches.

The source packets contain primary-paper text or the explicitly labelled metadata snippet. They do not contain Richmond's final conclusions, the old 47/40 reference, previous correspondence reports or external comparison judgments. The developer has studied the comparator; consequently this remains a development experiment, not a pristine unseen benchmark. In addition, model pretraining may contain the public review. Prompt isolation cannot rule that out.

2. Extract and inspect findings

For each paper, the extraction model proposes structured findings from the complete available text. Zero findings are allowed. Separate findings preserve differing contexts, comparators and follow-ups. A deterministic locator checks the entire quotation. A citation-repair role selects numbered page/line ranges; code copies those source lines while preserving the claimed component. It cannot rewrite the substantive finding. Missing, invalid or too-short source ranges retain the original unlocated quotation and fail the evidence gate. Raw model proposals remain in the call archive.

A separate critic receives the full available source and the extraction. It assesses every finding, including inference status, direction and qualification. Coverage is checked in code; omitted or duplicate finding indices are not silently accepted. The initial machine eligibility rule is:

```
eligible(f) = all_required_quotes_located(f) AND critic_verdict(f) == supported.
```

Here “supported” means the finding is defensible with its stated uncertainty and inference labels; it need not describe a fully observed causal chain. This conservative gate can sacrifice recall. Findings assessed partial, unsupported or uncertain remain in the evidence matrix for human inspection instead of disappearing. A high pass rate is not an evaluation target.

3. Construct the evidence graph

Each finding is a distinct configuration node connected to its context, resource, response and outcome nodes and their source-evidence nodes. No semantic canonicalization merges opposite concepts. The same lexical label in different findings remains separately attributable. The configuration itself represents the conditional relationship; evidence edges express attribution rather than experimentally established causation.

A separate undirected projection connects findings for retrieval. For token sets τ_i and τ_j , its weight is:

$w_{ij} = \text{size}(\text{intersection}(\tau_i, \tau_j)) / \text{size}(\text{union}(\tau_i, \tau_j))$, provided at least three retained terms overlap.

This Jaccard edge means “candidate for joint inspection.” It does not mean equivalence, corroboration, contradiction or a causal link. Stopword removal and lexical similarity can miss synonyms and connect misleading near-neighbours. All original text and polarity remain available to the synthesis model and reviewer. The threshold is a transparent engineering parameter, not an empirically optimized scientific constant.

The graph projection is partitioned with seeded Louvain modularity optimization, followed by splitting disconnected subsets. The modularity objective has the conventional form $Q = (1/(2m)) \sum_{i,j} [A_{ij} - \gamma k_i k_j / (2m)] I(g_i = g_j)$, with the implementation default resolution. This organizes evidence packets. It does not discover “true mechanisms,” and the number of communities is not the number of Richmond contexts.

4. Synthesize and return to primary sources

Each community produces a provisional, cited explanatory summary. A reduction stage considers the community summaries together with eligible findings and proposes programme theories. Every theory must cite existing finding identifiers. Each contains a conditional explanation, rival explanations, limitations and evidence gaps. No fixed number or names of context groups are supplied from Richmond.

Theories then generate source-search queries from their contexts, resources, responses and gaps. Page retrieval uses BM25:

$$\text{score}(q,d) = \sum_{t \in q} \log(1 + (N - \text{df}(t) + 0.5)/(\text{df}(t) + 0.5)) * \text{tf}(t,d)(k_1+1) / [\text{tf}(t,d) + k_1(1-b+b|d|/\text{avgdl})].$$

The configured values are $k_1=1.5$, $b=0.75$, with the top eight matching pages per query. Positive-scoring results retain paper and page identity. A bounded refinement pass re-examines these pages, the initial explanations and their underlying findings. New observations without an extracted finding remain gaps requiring extraction/review; they cannot acquire an invented finding identifier.

This is an explicit, restricted operationalization of retroduction: propose an explanation, seek evidence that could qualify it, revisit the source and refine the explanation. It is not a claim to recover a model's hidden internal reasoning, nor a theorem proving the hypothesis. The recorded explanation is a research rationale that can be criticized. One pass does not establish theoretical saturation.

5. Freeze production output before comparison

Write the final provisional programme theory and record its hash. Only then load the versioned Richmond reference into the evaluation stage. The comparison stage may identify differences but cannot edit the frozen system theory. Any later modification informed by the comparator starts a new, explicitly labelled development iteration.

Verification: the answer to “how?”

Verification requires three linked questions. First, **is the system's claim supported by its sources?** Second, **does its conditional explanation correspond to the published human synthesis?** Third, **do independent experts consider the explanation coherent, useful and appropriately qualified?** None of these questions can be replaced by a count of graph nodes.

The comparison matrix has one row per versioned reference configuration. It shows Richmond's context, resource, response and outcome beside the actual system explanation and its evidence identifiers. Six dimensions are judged: context, resource, response, outcome, direction and qualifiers. Qualifiers include population, comparator, time and whether an outcome or mechanism was observed or inferred.

Verdict	Operational meaning
Equivalent	The same conditional explanation is preserved across all six dimensions, with an actual system-theory counterpart and attributable evidence.
Partial	Some explanatory content is recovered, but at least one material component or qualification is absent, narrower, broader or changed.
Contradictory	Incompatible claims concern sufficiently comparable conditions and outcome definitions. Different contexts, follow-ups or comparators do not automatically qualify.
Not recovered	No substantive counterpart appears in the system synthesis. A relevant source finding alone is insufficient if the synthesis never used it.
Uncertain	Evidence availability, interpretation or reference ambiguity prevents a defensible decision.

AI verdicts are explicitly provisional. Code downgrades an “equivalent” verdict if a dimension is non-equivalent, if no system theory/evidence is cited, or if the cited findings are not linked to that theory. This removes certain internally inconsistent verdicts; it does not establish that the remaining semantic judgments are correct.

Two human reviewers should first ratify or amend the reference decomposition, then independently inspect the original paper and the system outputs using blank coder forms. They should not view the AI verdicts before their independent judgments. A later meeting adjudicates disagreements and records the reason, evidence and time spent. Reviewers should also assess system theories absent from Richmond, since a defensible new explanation is not necessarily an error.

The implemented exporter produces a separate `blind_review.html` without AI comparison verdicts, coder A/B CSVs and an adjudication CSV. Re-export preserves existing forms. The review reader requires identifiable reviewers, dates and rationales, and refuses adjudication without both initial ratings. It computes paired agreement only for completed rows and whole-reference recovery only after every reference row is adjudicated. These controls validate record completeness; they cannot authenticate a person's identity or substitute for actual independent review.

After complete adjudication, exact configuration recovery is $\text{number equivalent} / \text{number reference configurations}$. Report partial, contradictory, not-recovered and uncertain counts separately. Do not silently count partial as exact or assign an arbitrary half-credit score. If only a subset is reviewed, report the reviewed fraction and the sampling design; do not label the subset result whole-corpus recall.

Source-support precision is a different measure: $\text{supported system claims} / \text{reviewed system claims}$, using independently adjudicated claim-level support. It requires review of generated content, including content that has no Richmond counterpart. Precision cannot be inferred from reference recall. Automated quote-location rate is $\text{located full quotations} / \text{all proposed quotations}$, and must be labelled as such.

For paired categorical human judgments, Cohen's kappa is $(p_o - p_e) / (1 - p_e)$, where p_o is observed agreement and p_e is chance agreement calculated from the raters' marginal distributions. Report raw agreement, category counts, disagreements and denominators alongside kappa. With no paired human

judgments, kappa and human recall/precision are **not available**, rather than zero or an AI estimate relabelled as human agreement.

Architecture and tool decisions

The immediate architecture is a source layer, typed finding layer, evidence-graph layer, synthesis layer and separate evaluation layer. Named model roles have explicit inputs and outputs; deterministic Python controls sequencing, schema checks, persistence and budget gates. Different roles do not require different foundation models, and different model names do not establish independent human judgment.

Component	Current decision	Reason and limit
Language and contracts	Python 3.11, Pydantic	Fits the existing repository and provides explicit validated data contracts. Schema validity is not factual accuracy.
Model access	OpenAI SDK with structured JSON-schema outputs	Record exact requests and returned model identity. API success is separate from scientific success.
Extraction and synthesis	GPT-5.5 snapshot gpt-5.5-2026-04-23, medium reasoning	Retains the established strong-model choice while making it reproducible; no claim it is optimal for this task.
Citation repair, critic and comparator	GPT-5.4 mini snapshot gpt-5.4-mini-2026-03-17, medium reasoning	Limits cost for repetitive audit tasks; semantic verdicts still need expert assessment.
PDF parsing	pypdf, page-preserving text plus original PDFs	Keeps source addresses; layout and extraction defects must be inspected.
Evidence graph	NetworkX and explicit JSON graph	A small corpus can be inspected without a graph server. Lexical grouping is a baseline, not a validated ontology.
Retrieval	BM25 primary-page retrieval and graph-community evidence packets	Deterministic and inspectable; synonym recall is a limitation requiring measured comparison with embeddings.
Run persistence	Immutable request/response files, JSON/JSONL manifests and CSV/HTML outputs	Isolates the experiment from the mutable legacy PostgreSQL database. Existing PostgreSQL/Neo4j artifacts are preserved.
Human-facing output	HTML report, evidence CSV, theory CSV, comparison CSV and separate coder forms	Readers can follow a conclusion to a source without querying a database or understanding graph internals.

The official model documentation lists structured-output support for GPT-5.5 and standard input/output rates of USD 5/30 per million tokens; GPT-5.4 mini lists USD 0.75/4.50. The budget guard reserves a conservative per-call upper estimate and records actual token-based estimates. Unresolved calls retain a reservation; automatic retries are disabled to avoid silently duplicating uncertain charges.^[13]

This implementation is a **custom graph-assisted synthesis experiment**, not a drop-in implementation of Microsoft's full GraphRAG indexing and query stack. It does not claim that a citation/co-citation network,

vector embeddings, link prediction, domain fine-tuning or RLHF has been implemented merely because corresponding packages or proposal paragraphs exist. Domain adaptation in this run is prompt/schema adaptation. The accepted proposal's stronger training claims need either actual training experiments or an explicit methodological revision agreed with the professor.

For a later production system, preserve the same evidence contracts and add genuine human gates and versioned feedback before downstream approval. Graph-guided extraction, approved concept merge/split operations, multiple refinement cycles and resumable human intervention are substantive extensions to evaluate; they should not be claimed complete because a simple workflow can run without stopping. The exploratory machine run keeps every human gate pending and cannot grant itself research approval.

September 23 clarification: semantic relations and two levels of comparison

The July 24 professor meeting explicitly requests concept/relation evaluation and results before and after human intervention. The September attribution graph and 18-row configuration reference do not complete those requirements. The historical 47 concepts/40 relations remain an evaluation track requiring source reconciliation; they must not be silently discarded or treated as an already authenticated gold dataset.

The implementation must distinguish four objects:

- An **entity mention** is an attributable occurrence of a concept or construct, with its role, paper and source span. A **canonical concept** is an explicitly justified grouping of mentions, not a synonym inferred from a shared role.
- A **semantic relation assertion** describes what connects particular entities under particular conditions. It needs evidence for the relationship, not merely one quotation mentioning each endpoint.
- An **attribution edge** such as `has_context` or `cites` records where a component or quotation belongs.
- A **retrieval edge** such as `lexical_retrieval_candidate` selects potentially relevant records. It does not assert that their explanations agree.

The new offline inspection exporter exposes existing instances, relations, configurations and attribution/retrieval edges separately. It preserves the legacy labels, quotes and critiques; it does not repair the scientific content by displaying it. Its source hashes and endpoint checks establish data identity and referential integrity, not semantic validity.

Research basis and extension contract

Dagdelen et al. evaluate joint entity/relation extraction with structured, potentially hierarchical records. Their examples show that losing parts of a compound relation can change its scientific meaning. Their manual evaluation allows equivalent information despite wording differences, while still checking the grouping of entities. Their experiments concern materials science and task-specific fine-tuning; they do not validate education-domain realist synthesis or our current prompting baseline.^[14]

Our resulting design decision is to preserve a conditional assertion as a first-class record:

```
a = (assertion_id, entity_ids, predicate, context, outcome_definition, comparator, timepoint,
evidence_ids, inference_status, limitations).
```

This is a data contract, not a numerical causal model. Semantic extraction must be source-led, allow partial assertions, and decline an unsupported link even when the endpoint types permit it. A graph edge may point to this assertion record; it cannot replace its qualifiers. Source-reported associations, primary-author explanations and synthesis hypotheses require separate labels. Candidate semantic merges must preserve role, polarity and scope, with reversible merge/split records. Benchmark labels and judgments remain external to production prompts.

Implementation status: typed semantic assertion contracts and offline source/structural checks are implemented, but semantic extraction under this extension has not run. Existing configuration records already retain many of these qualifiers. Existing `has_*` edges must not be relabelled as an implementation of the extension. The completed baseline supplies actual extraction and synthesis behavior for inspecting failures before the extension is evaluated.

Evaluation contract

Keep three reference-facing tables: concepts, relations, and complete conditional explanations. For a relation, the reviewer checks both endpoint meanings, the asserted connection, direction, context, comparator, time and inference status. Exact string identity and semantic equivalence are separate columns. Do not award an exact semantic match merely because both endpoints appear somewhere in the graph.

For generated-output precision, review the generated assertion inventory, including assertions without a Richmond counterpart. For reference recovery, review the ratified reference inventory. These denominators differ. Redundant generated paraphrases must not inflate recovery; relation matches must preserve the connected assertion rather than pool endpoints from unrelated studies. Report partial matches separately from equivalent matches.

Before/after human intervention is a versioned experiment: freeze the automatic result, record reviewer edits with evidence/reasons/time, execute only affected downstream stages in a new version, and reevaluate with the same rubric. Changes informed by Richmond are development corrections; they are not a held-out generalization result. A corrected reference also requires a new reference version and transparent recomputation of both comparison conditions.

RAMESES requires changes from the intended method to be described and justified. Accordingly, missing proposal components (including model training and link prediction) remain explicit pending requirements or require a documented revision agreed with the professor; a simpler implemented baseline cannot silently redefine them as achieved.^[2]

Experiments needed for a publishable effectiveness claim

Further September 23 work is specified in `SEMANTIC_ASSERTION_METHOD.md` (source-specific entity/assertion contracts, comparator/time and direction-versus-benefit distinctions, and additional primary extraction research) and `EVALUATION_PROTOCOL_V2.md` (strict theory-level comparison of the frozen baseline). Contracts and structural tests are implemented; the semantic extraction experiment is not executed. Evaluation v2's first API request was rejected for exhausted credit. Its planned outputs must not be substituted for observed baseline results.

The current run can establish that the output exists, is traceable, and can be evaluated with a specified protocol. It cannot by itself establish superiority of multiple agents or GraphRAG. A defensible comparison should hold corpus, model, output schema and budget constant while comparing: direct source-based

synthesis; extraction plus source retrieval; and extraction plus graph-organized retrieval and refinement. A separate ablation removes the critic to measure both error reduction and evidence loss.

Reviewers should judge anonymized system variants and the same source corpus using a fixed rubric. Repeated runs quantify stochastic variation. Report cost, review time, recovery, unsupported claims, explanatory completeness and useful departures. Paper families, rather than every extracted node, are the appropriate units for assessing dependence when designing resampling or sampling plans. A small worked example warrants cautious generalization.

Predeclare error categories: unavailable text, parsing error, missed evidence, incorrect context/resource classification, inference inflation, polarity/comparator/time distortion, unsupported cross-paper linkage, synthesis omission, retrieval omission, and ambiguous reference. Investigate disagreements before deciding whether the benchmark, system or both need revision. This is how a result becomes useful research evidence rather than merely a higher score.

Sources

- [1] Private project sources: documents/abstract-from-professor/abstract.md; January 23, February 4, February 19, May 28 and July 24 meeting transcripts in transcript-meetings/; professor briefing documents in documents/. These establish intended scope, not empirical performance.
- [2] Wong G, Greenhalgh T, Westhorp G, Buckingham J, Pawson R. RAMESES publication standards: realist syntheses. *BMC Medicine*. 2013;11:21. Especially items 8-14 and discussion of inferences and reporting.
- [3] Dalkin SM et al. What's in a mechanism? Development of a key concept in realist evaluation. *Implementation Science*. 2015;10:49. Sections on disaggregating resources/reasoning and continuums of activation.
- [4] Edge D et al. From Local to Global: A GraphRAG Approach to Query-Focused Summarization. Version 2, 2025. Also supplied as documents/paper-Microsoft-GraphRAG.pdf. This source evaluates summarization, not the present realist task.
- [5] Skarlinski MD et al. Language agents achieve superhuman synthesis of scientific knowledge. 2024 preprint. Methods and human-evaluation sections; the title is the authors' claim about their tasks.
- [6] Rouzrokh P, Shariatnia M. LatteReview: A Multi-Agent Framework for Systematic Review Automation Using Large Language Models. 2025 technical report. Workflow and evaluation sections. Table 2 and accompanying threshold descriptions contain ordering/reporting inconsistencies; numerical results are not imported as expected performance here.
- [7] Madaan A et al. Self-Refine: Iterative Refinement with Self-Feedback. 2023. Feedback-based inference-time revision, rather than weight training.
- [8] Huang J et al. Large Language Models Cannot Self-Correct Reasoning Yet. 2023/2024 preprint. Scope is intrinsic correction on the reasoning tasks studied.
- [9] Wadden D et al. Fact or Fiction: Verifying Scientific Claims. EMNLP 2020. Scientific claim stance and evidence rationale task.
- [10] Min S et al. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. EMNLP 2023. Atomic factual support; primarily evaluated on biography generation.
- [11] Traag VA, Waltman L, van Eck NJ. From Louvain to Leiden: guaranteeing well-connected communities. *Scientific Reports*. 2019;9:5233. NetworkX Louvain documentation, accessed September 2026.
- [12] Ren Z et al. Systematic Literature Reviews With Two Multi-Agentic Systems And Human-In-The-Loop. July 2026 preprint; discovery-stage source, abstract inspected only, not used to substantiate implementation effectiveness.
- [13] OpenAI, GPT-5.5 model documentation, GPT-5.4 mini model documentation, and Structured outputs, accessed September 2026. Prices are configuration assumptions checked against the official documentation; account billing remains authoritative.
- [14] Dagdelen J et al. Structured information extraction from scientific text with large language models. *Nature Communications*. 2024;15:1418. DOI 10.1038/s41467-024-45563-x. Introduction/Figure 1 and exact-match versus manual evaluation discussion. Revisited September 23, 2026; the publisher page failed to open, so the full-text PMC version was used. This is supporting research for an extension, not evidence that the extension already performs well.

Source-led conditional semantic assertions

Decision and research basis

Keep the September baseline intact. Extend the production representation with separately identifiable source entities and conditional assertions. Do not relabel `has_context`, `cites` or lexical-similarity edges as extracted semantic relations. This extension is source-led; Richmond's entity/relation inventory remains external evaluation material.

Nye et al. separate evidence extraction, entity linking and reported-result inference. They emphasize intervention/comparator/outcome combinations and document-level context, and report errors when decreasing an undesirable outcome is confused with a negative effect. Their clinical-trial experiments are not validation of educational realist inference. The useful design implication here is to keep comparator, outcome definition and statistical direction explicit, and distinguish magnitude direction from educational benefit/harm. Their pipeline also illustrates error propagation; competent modules do not guarantee correct end-to-end output. Primary paper, sections 1.1, 2.2 and 3.2.

Luan et al. model entity spans, relations and coreference jointly in SciIE and introduce SciERC. The annotation domain is AI research abstracts, not realist educational evidence. The paper's knowledge-graph construction selects a most-frequent relation between an entity pair. We reject that aggregation rule for this project: different conditions can legitimately yield different relations, and minority adverse/null evidence must remain visible. Their distinction between mentions, relations and coreference informs our separate records; their labels and trained scores are not imported. Primary paper, sections 3-5.

DyGIE++ refines contextual span representations through learned graph propagation for information extraction. Those graph updates are part of a trained extraction model; they are not equivalent to GraphRAG retrieval over our stored evidence graph. Its methodology motivates testing contextual extraction rather than isolated keyword pairs, but does not establish that installing its code or copying a graph schema provides domain adaptation. Primary paper, section 2.

Dagdelen et al.'s structured extraction work and the realist resource/response distinction remain additional foundations described in `METHOD_AND_VERIFICATION_PROTOCOL.md`. The contract below is our design inference from those methods and observed baseline failures, not an algorithm claimed verbatim from any one paper.

Representation

An entity record identifies a source-specific mention or a labelled source-grounded abstraction, its role, original quotations and page locators. Roles cover context, educational resource, learner response, outcome, population, assessment and study design. An abstraction is not a verbatim named-entity occurrence and must not be counted as span-level NER success. No cross-paper synonym merge is automatically approved by this contract.

A conditional assertion is

```
a = (paper, subject, predicate, object, kind, context, comparator, time, outcome_definition,
statistical_direction, educational_interpretation, inference_status, relation_evidence, limitations).
```

The graph may display a subject-predicate-object edge pointing to *a*, but the edge does not replace the assertion's qualifiers. The same endpoint pair may have several assertions for different conditions or times. Relation evidence is checked independently from quotations locating the endpoint entities. A quote containing both endpoints is not sufficient proof of the relationship.

`statistical_direction` distinguishes increase, decrease, no statistical difference, mixed, unmeasured and not applicable. `educational_interpretation` distinguishes benefit, harm, no detected difference, mixed, unclear and not applicable. A null statistical contrast cannot automatically become harm or benefit. An author explanation or model hypothesis cannot be labelled as a measured contrast; a compound sentence mixing the two must be split into separate assertions. Missing context/comparator/time must be explicit rather than invented.

Intended extraction and reasoning sequence

1. Read a page-preserving source packet and identify relevant source spans and constructs, including null/adverse findings and partial configurations. Return entities and assertions jointly so the relation refers to actual local entity IDs.
2. Locate each quotation against the frozen source; preserve exact source spans and any unresolved quotations. Check endpoint identity, duplicate IDs, source availability and direction/inference consistency.
3. Independently assess whether the source supports the asserted connection and qualifications. A located quotation is not a pass on this semantic assessment. Keep rejected/uncertain candidates visible.
4. Propose reversible concept equivalences with explanations; preserve mention identities, role and scope. A shared word, entity type or embedding similarity supplies a candidate, not an accepted semantic merge.
5. Retrieve complete assertion neighborhoods with their conditions and provenance for synthesis. Keep source-reported connections distinct from cross-paper hypotheses. Never promote a predicted missing edge into observed evidence.
6. Evaluate source-span extraction, semantic assertions, global reference recovery and programme-theory correspondence as separate tasks. Record failures at the earliest attributable stage and preserve pre-intervention output.

This sequence is an extension to evaluate, not a statement that every step is implemented.

`semantic_contracts.py` currently implements the typed records and offline source/structural audit. Its tests cover missing relation evidence despite located endpoints, dangling endpoints, duplicate identifiers, null-versus-harm mistakes, and inference inflation. No actual semantic extraction, trained entity model, new canonical inventory, relation critic, learned embeddings or link prediction has run under this extension yet.

Evaluation denominators

The 47/40 project reference concerns recovery of constructs and relationships in Richmond's synthesis. It is not a fully annotated source-span corpus and therefore cannot by itself supply supervised NER/RE precision/recall. Span-level NER evaluation needs source-span annotations. Semantic relation precision needs human judgments on generated assertions, including assertions absent from the Richmond reference. Reference recovery needs a ratified reference and must not combine endpoints from unrelated studies to manufacture a match.

For relation correspondence, compare endpoint meanings, predicate, context, comparator, time, outcome definition, direction and inference status. Report exact strings separately from semantic equivalence and partial correspondence. For a benefit with decreasing outcome magnitude, compare the actual outcome meaning before assigning polarity. Neither an exact anchor check nor passing the new contract tests establishes these judgments.

The next executable extension experiment should pilot joint source extraction and source auditing on contrasting papers, then inspect errors before full-corpus execution. Pilot selection and any changes after reading Richmond are development choices. Effectiveness remains to be tested against simpler source-based and retrieval baselines with a consistent schema, model/budget and independent human rubric.

Frozen-theory correspondence evaluation v2

Reason for a new evaluation

The September 23 baseline's four partial rows RC04/RC05/RC11/RC12 use findings outside the selected theory, including one row selecting no theory. The first validator applied evidence-link checks only to full equivalence. V2 evaluates the same frozen production result under a stricter, explicit theory-level contract. It does not alter the original 180 findings, nine theories, reference, or earlier comparison responses.

This protocol is development work informed by observed benchmark/evaluator behavior. A change in labels is an evaluation-method change, not improvement in the production system and not held-out validation. The 18-branch task remains separate from concept/relation recovery and source-support assessment.

Pre-execution specification

Run `res_pipeline.evidence.evaluation` against `outputs/runs/evidence-20260910-full`, writing a sibling `outputs/runs/evaluation-20260923-v2`. Evaluate all 18 reference rows, not only the four known problem rows. Use the existing pinned comparator model and pricing configuration. The budget includes this evaluation, the source run, its unresolved reservations and the separate pilot; the existing USD 50 ceiling is retained. Prior AI comparison labels and source-critic verdicts are not sent in new requests.

For each reference, the model sees the complete frozen programme theory and the scientific content of findings linked to those theories. It must select actual theory IDs and supporting findings already linked to those selected theories. Findings unrelated to the selected theories cannot count as theory recovery. It assesses context, resource, response, outcome, direction and qualifiers once each. Each equivalent/partial dimension must include a verbatim anchor to a named string field of a selected theory.

An equivalent, partial or contradictory verdict requires a selected theory and linked evidence. Equivalent also requires all six dimensions equivalent. Partial requires substantive dimensional correspondence. Not recovered means no selected counterpart and no claimed equivalent/partial dimensions; uncertainty is recorded separately. A broad theory with mixed outcomes may contain a matching conditional branch, so the evaluator must inspect that branch instead of comparing only the overall direction field. An additional defensible qualifier is not automatically a contradiction.

The validator rejects unknown/duplicate IDs, findings outside selected theories, missing/duplicate dimensions, inconsistent verdict prerequisites and unmatched/misassigned quotation anchors. Failed rows are **quarantined**, preserving the raw scientific verdict; no automatic relabelling, score imputation or silent prompt retry is performed. Successful admission means only that these scope/location checks passed. A model could cite an exact but semantically irrelevant passage, so admission is not entailment or human verification.

Outputs and audit

The separate directory preserves exact request/response/parsed records, token usage, actual model identities, configuration, code/prompt/input hashes, row admission reasons, JSON/CSV comparison records and a readable HTML matrix. Source files are hashed again after evaluation to detect mutation. The HTML links every theory anchor and supporting finding back to the frozen baseline report. Original evaluator labels may be displayed for audit but are not model inputs.

Independent reviewers still need to ratify the Richmond operational reference, judge source support and assess conditional correspondence. Their existing forms remain blank. The evaluation run does not authorize or impersonate human approval. Structural test fixtures cover the observed outside-theory partial-match failure and absent-theory partial verdict, false quote anchors, invalid no-counterpart matches and admissible control cases; these tests do not establish scientific validity.

September 23 baseline: interpretation and review priorities

Status and purpose

The source extraction/audit stage produced 180 findings from all 28 available-source records; 174 passed the machine gate. The frozen final synthesis contains nine provisional theory statements. These are generated outputs, not independently established scientific conclusions. Refer to `CURRENT_RUN_RESULTS.md` for terminal comparison/export status and costs.

The theory file SHA-256 is `d0489f641746eeb0389343e28c0db521e441e582ff98bf8fcfb1cb0ca81bdf7f`. The original is preserved in `outputs/runs/evidence-20260910-full/programme_theory.json`. Review observations below do not modify that file and are not human adjudication.

What the nine generated statements cover

ID	Subject of the machine-generated explanation
PT01	Case-proximal diagnostic prompts versus abstract or delayed counter-bias instruction
PT02	Early, concrete visual-diagnosis strategy instructions versus late or vague combined instructions
PT03	External diagnostic schemas, task fit and limits of generalization
PT04	Self-explanation/reflection, prior knowledge and immediate versus delayed transfer
PT05	Error exposure and feedback sufficient to explain or repair mistakes
PT06	Varied cases and virtual patients, separating observed learning from design rationale or popularity
PT07	Repeated key-feature questions, feedback, time-on-task and item-specific retention
PT08	Simulation/game responsibility, preparedness, stress, usability and expertise
PT09	Governance, topic fit and uptake of virtual-patient/EHR resources

These are primarily organized around instructional resources. Richmond organizes the final explanatory account around five overlapping learner contexts. A different organization is not automatically wrong, but the system must still recover or defensibly challenge the important conditional mechanisms. Neither the number of theories nor their fluent wording establishes this.

Concrete interpretation issues found during assistant inspection

1. **PT02 study independence is not established as written.** Its limitations say that S004 and S005 are related study families and should not count as independent replication. The manifest supplies distinct family IDs, and the supplied methods describe different designs/sample sizes. Shared authors and ECG tasks alone do not establish shared participants. Verify recruitment/sample overlap before accepting this warning; do not silently merge the studies or call the statement human-validated. S013/S020, by contrast, are explicitly linked as a secondary analysis in the supplied record.
2. **PT05 title states necessity more strongly than its evidence establishes.** “Only when” implies a necessary condition across error-based learning. Its own limitations acknowledge that S018 did not directly compare detailed feedback/coaching against diagnosis-only feedback, and that the prior-knowledge threshold is unknown. The body presents a hypothesis; an expert should decide how narrowly the headline can be supported. This is a candidate overclaim, not a revised output.
3. **PT04's S026 delayed claim needs its exact comparator.** S026-F01 reports within-control-group improvement after self-explanation, not a randomized superiority contrast against no self-explanation. PT04 summarizes this as delayed improvement from self-explanation alone. A reviewer must retain the within-group design and consider testing/time effects before treating it as an intervention effect. S026-F04 separately supplies between-group far-transfer contrasts. Excluded S026-F02/F03 remain available for review; a machine exclusion is not an expert judgment.
4. **All nine direction fields are mixed.** This is understandable for broad summaries containing benefits and failures, but too coarse to function as an atomic relation label. Relation evaluation needs separate assertions preserving the specific context, comparator, outcome and time. A single mixed-direction theory must not be counted as both a recovered positive and a recovered negative relation without those links.
5. **The implementation admits synthesis after an automated source gate.** A gate checks individual findings, whereas synthesis can still introduce new unsupported connections or generalizations. Therefore the nine final theories need their own source-support assessment. Source linkage and a valid finding ID alone do not establish entailment.
6. **Four partial comparison rows overreach their selected-theory evidence.** RC04 cites S005-F01/S013-F02/S016-F03 outside PT01; RC05 cites S021-F01 outside PT01/PT03/PT08; RC12 cites S005-F04 outside PT01; RC11 selects no theory but cites S028-F01/F08. The baseline validator enforced selected-theory linkage for full equivalence only, leaving these partial assessments unflagged. An independent post-run check found them. The original comparison is preserved; inspection v4 visibly flags them. A corrected evaluation must distinguish source-finding correspondence from final-theory recovery instead of pooling those levels. The raw 14 partial/four not-recovered distribution must not be reported as a validated recovery score.

A worked comparison, not a claimed human score

The provisional reference row RC01 operationalizes Richmond Figure 3's low-knowledge pathway involving near-peer examples with prompts, perceived similarity/feeling at ease, and improved learning or accuracy. The actual system counterpart PT04 discusses prompted resident examples and delayed transfer, with findings including S026-F04 and S025-F05. It does not establish the same perceived-similarity/at-ease mechanism. The saved AI comparator therefore marked RC01 **partial**.

This illustrates the required distinction: recovering an intervention and a favorable outcome is insufficient if the explanatory response is missing. The evaluator must also verify that the reference decomposition fairly represents Richmond and that the system's interpretation is supported by its primary sources. The AI label is an aid to inspection, not the human verdict or a measured accuracy rate.

Prepared human work

- `outputs/runs/reference-review-20260923-v1/index.html`: ratify 47 concepts, 40 relations and 18 configurations against Richmond. These are project-coded reference units, not a dataset published by Richmond. Separate A/B/adjudication forms contain blank decisions.
- `outputs/runs/source-review-20260923-v1/index.html`: assess source support of every extracted finding and every frozen theory, with critic/eligibility/comparison verdicts withheld. Six forms cover findings/theories for A/B/adjudication. Review both machine-included and excluded findings to observe false acceptance and false exclusion.
- The baseline run's final `blind_review.html` and `human_review_coder_A/B.csv` support the separate configuration-correspondence task after reference ratification. The 47/40 system-scoring track remains additional work; reference ratification forms alone do not complete it.

Reviewers have not yet been appointed. No human cells are filled, no human agreement or precision is inferred, and no after-human-intervention improvement is claimed. Subsequent corrections must be attributed and stored in a new output version, preserving this pre-intervention baseline. Review packets should be completed independently before reviewers see AI verdicts or this assistant's issue list.

Method implications

The baseline provides actual inspectable candidates and testable failure modes. The next implementation should preserve atomic semantic assertions separately from broad theory summaries, link each assertion to its supporting finding(s), and distinguish observations from cross-paper explanatory hypotheses. It must evaluate entity/relation recovery and source support separately from whole-configuration correspondence. Changes informed by these Richmond comparisons are development work, not held-out validation. The accepted proposal's unimplemented training, link-prediction and human-intervention components remain explicit in `REQUIREMENTS_AND_ACCEPTANCE.md`.